UNIT 1 · TOPIC 3(E)

Transplanting Equipment

Syllabus topic: Equipment for land development, tillage, sowing, planting and transplanting

Self-propelled rice transplanter

- Needs a **mat-type nursery** — seedlings raised on a soil layer over polythene in trays.
- Fixed row spacing **238 mm (≈23–24 cm)** in Japanese machines (30 cm models also exist).
- Hill spacing adjustable **12–21 cm**; **2–4** seedlings per hill.
- A finger/fork picking mechanism is driven by a four-bar or planetary linkage.
- Walk-behind (**4–6 rows**) and riding (**6–8 rows**) types.
- Field capacity ≈ **0.15–0.2 ha/h** (6-row).

Vegetable transplanters (semi-automatic)

- The operator feeds seedlings/plug seedlings into rotating cups or fingers.
- Automatic types need **pro-tray (plug) seedlings**.

Practice MCQs

Q28. Mat-type nursery is a prerequisite for:

(a) Potato planter
(b) Self-propelled rice transplanter
(c) Sugarcane cutter planter
(d) Pneumatic planter

Q29. Standard row-to-row spacing of Japanese-type mat rice transplanters is:

(a) 150 mm
(b) 200 mm
(c) 238 mm
(d) 300 mm

Answers

Q28 — (b) Self-propelled rice transplanter

Q29 — (c) 238 mm